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Studierichting: Informatica
Oefeningenreeks 1B nr 37.2

$$-3-4i=(x+yi)^2=x^2-y^2+2xyi$$

$$\begin{cases} x^2-y^2 & = & -3 \\ 2xy & = & -4 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2-y^2 & = & -3 \\ x^2(-y)^2 & = & -4 \end{cases}$$

$$\Rightarrow \lambda^2+3\lambda-4=0$$

$$D=9-4\cdot 1\cdot(-4)=35$$

$$\lambda_1=\frac{-3+5}{2}=1=x^2\Rightarrow x=\pm 1$$

$$\lambda_2=\frac{-3-5}{2}=-4=-y^2\Rightarrow y=\pm 2$$

$$\Rightarrow (x+yi)\in\{-1+2i,1-2i\}$$

References